# PA12 coated copper strip

## Bendability for the benefit of electric vehicles

Luvata's metallurgical, manufacturing and insulation expertise continues to support the future of EVs.

Through our continuous improvement, Luvata has succeeded in developing a high-adhesion PA12 coated copper strip.

Luvata PA12 coated copper strip is specifically designed for the growing EV market. It is able to withstand voltage of nearly 30kV and commonly used for battery connections.

Luvata PA12 coated copper strip is offered in a very soft temper, so it is easy to form and bend. The product can be bent into sharp angular shapes and even loops, meanwhile the adhesion of the insulation remains tight.

Luvata standard grade oxygen-free copper (Cu-OF) is a high purity copper that is immune against hydrogen embrittlement. It is used in applications where high electrical and thermal conductivity are essential.

It can be easily joined using all welding and brazing methods and is ideal for manufacturing processes requiring extreme deformability.

**About Luvata**

Luvata is a world leader in metal solutions manufacturing and related engineering services to industries such as renewable energy, automotive, healthcare, and power generation and distribution. The company's continued success is attributed to its longevity, technological excellence and strategy of building partnerships beyond metals. Employing approximately 1,400 staff in 7 countries, Luvata works in partnership with customers such as ABB, CERN, Siemens and Toyota. Luvata is a group company of Mitsubishi Materials Corporation.

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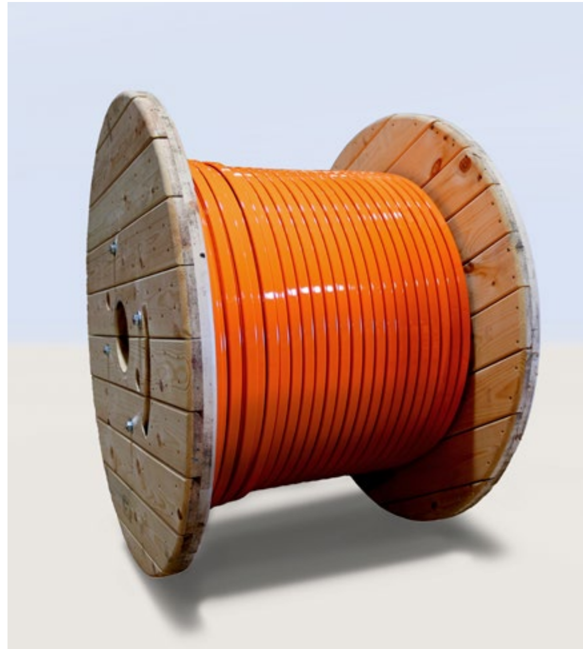
**Technical specifications**

| | |
|---|---|
| Alloy | standard grade oxygen-free copper (Cu-OF) |
| Dimensions | width 8-20mm, thickness 2-6mm, full round edges |
| Soft temper | R200 |
| Adhesion | excellent coating adhesion |
| Flammability rating | tested according to UL 94 HB |



Luvata PA12 coated copper strip on wooden reel.

**Packaging**

Luvata high-quality PA12 coated copper strip is cut to the exact length of customer specifications and delivered on reels.

**Support in overcoming the challenges facing the e-mobility industry**

Luvata is focusing on copper's high thermal conductivity in designing next generation products that overcome customers' challenges in terms of charging times, heating, cooling, driving ranges and infrastructure.

Luvata recognizes these challenges as opportunities and has made improvements specifically targeting the accelerated pace of e-mobility research and development by offering:

- Custom designed semi-products
- Insulated semi-products or components ready for assembly
- High performance alloys
- Specifications with extreme tolerances
- Accelerated prototype manufacturing

Luvata PA12 coated copper strip available in all RAL standard colours.

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# Data Sheet

GENERAL DESCRIPTION
– SUBJECT TO CHANGES OR DEVIATIONS

## Oxygen-free Copper Cu-OF standard grade – Luvata Alloy Cu-OF

**Alloy description**

Luvata Cu-OF standard grade oxygen-free copper is high purity copper that is immune against hydrogen embrittlement. It is used in applications where high electrical and thermal conductivity are the essential requirements. It can be joined with all welding and brazing methods and it is suitable for manufacturing processes requiring extreme deformability.

**Typical applications:**

- Electrical components
- Switchgear applications
- Generator material in rotor and stator windings
- Other applications where high electrical and thermal conductivity is needed

**Products / shapes:**

Round rods, rectangular bars and solid profiles. Corresponding EN-norms for different products are as follows:

- EN 13601 – Copper and copper alloys.
  Copper rod, bar and wire for general electrical purposes.

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**Chemical composition and corresponding standards:**

| Luvata Pori Oy alloy | Composition * % | EN – CEN/TS 13388:2008 | ASTM / USA |
|---|---|---|---|
| Cu-OF | Cu + Ag min. 99,95 % | Cu-OF / CW008A | CDA C10200 |

* Other elements max %: O 0,0010,Bi 0,0005, Pb 0,005

**Physical properties:**

| Density kg/dm$^3$ | Coefficient of linear expansion 1/K | Specific heat J/(kg x K) | Melting temperature °C |
|---|---|---|---|
| 8,94 | 0,0000177 | 385 | 1083 |

**Mechanical properties – typical values:**

| | Soft temper | Half-hard temper | Hard temper |
|---|---|---|---|
| Hardness HV | 35 – 65 HV | 70 – 95 HV | 85 – 115 HV |
| Tensile strength | 200 – 220 N/mm$^2$ | 250 – 350 N/mm$^2$ | 260 – 400 N/mm$^2$ |
| 0,2% yield strength | 35 – 65 N/mm$^2$ | 180 – 280 N/mm$^2$ | 220 – 380 N/mm$^2$ |
| Elongation | min. 40 % | min. 12 % | min. 5 % |

**Electrical and thermal properties – typical values:**

| | | | |
|---|---|---|---|
| Electrical conductivity | vol | % IACS * | min 100 |
| | mass | %IACS | min 99,4 |
| | MS/m | | min 58,0 |
| Electrical resistivity | vol | Ω mm$^2$/m | max 0,0172 |
| | mass | Ω g/m$^2$ | max 0,1533 |
| Thermal conductivity (20 °C) | W / Km | | 390 |

* % IACS = International Annealed Copper Standard. The % IACS values are calculated as percentages of the standard value for annealed high conductivity copper as laid down by the International Electrotechnical Commission.

**Joining and machining:**

| Machinability rating (free cutting brass = 100) | Soldering | Brazing | TIG | MIG | EBW |
|---|---|---|---|---|---|
| 20 | Excellent | Excellent | Good | Good | Good |

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# Oxygen-free Copper

Luvata's highly malleable and consistent oxygen free copper offers superior electrical and thermal conductivity.

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# The Purest Commercial Copper Grade

The purity of oxygen-free copper makes it exceptionally conductive. This quality, along with its malleability and metallurgical consistency, makes it ideal for the most precise industrial and scientific applications. It can be found in superconducting wires, circuit boards, vacuum interrupters, X-ray tubes, laser mirrors and scientific instruments, where precision is critical, and the copper needs to be of a consistently high quality.

**Purity is only part of the story**

It is the purity of oxygen-free copper that makes it superior to other commercial copper alloys, but the people behind the product are equally important. Whether it is a case of needing high electrical conductivity, a material that resists deformation in the manufacturing process, or secure joining by brazing, our people have the experience and insight to know when and how oxygen-free copper can be used to its best advantage. We believe in forging strong, long-term relationships and engaging fully with our customers' businesses, helping them solve problems, innovate and design tailor-made solutions

Our in-house casting allows us to fully control the production process and the quality of our end products. We have just invested in a state-of-the-art hot extrusion line in Finland including the latest in material handling robotics and automation, allowing for even greater product customization.

*"As we all strive towards a greener, more digitized future, oxygen-free copper will play an increasingly important role. It is used in scientific applications such as fusion research for renewable energy, and can be found in the superconductors at the CERN particle accelerator. It is also central to the green transition, through the production of renewable electricity and the development of e-mobility solutions. In fact, we see unlimited potential for the future of oxygen-free copper in building a more sustainable future, including solving challenges that haven't even been thought of yet.*

*We are serious about sustainability. Every strategic decision we make takes into account our environmental impact, and we have pledged to become carbon neutral by 2045. We believe oxygen-free copper as fundamental to our vision of a future which benefits people, society and the Earth."*

President & CEO
**Pekka Kleemola**

# Luvata OF-OK® Copper

## Luvata Oxygen-Free Copper was specifically developed for electrical, electronic and heat transfer applications. It is an extremely pure copper alloy, with a maximum oxygen content of 5 ppm (0.0005%) and has a unique combination of high-performance properties.

**Consistency**
High quality raw materials, melting and casting in electric furnaces, and a strictly controlled manufacturing process, guarantee the material is consistent from batch to batch.

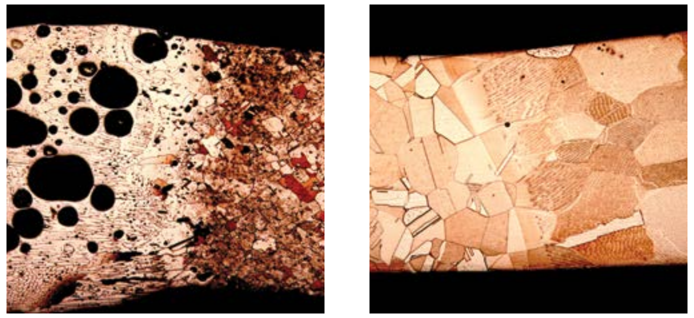
Consistency means less time consumed for machine set-up and less waste. By using silver bearing oxygen-free copper (alloying copper with a small amount of silver) its softening temperature can be increased significantly without compromising its high conductivity or formability. Silver bearing Luvata OF-OK® (Ag) is perfectly suited to commutators, generator rotor bars and welding components.

**Electrical and Thermal conductivity**
As a result of its purity, Luvata OF-OK® copper has high electrical conductivity (typically 100 – 102% IACS) and high thermal conductivity (390 W/Km). The higher conductivity values compared with other materials mean that equipment made of Luvata OF-OK® copper operates more efficiently and at lower temperatures, which increases their service life and reliability.

**Immunity to Hydrogen embrittlement**
At high temperature, hydrogen ions can diffuse into copper and form water with the oxygen atoms in the copper. The resulting water vapor has a high enough pressure to form pores in the copper known as hydrogen embrittlement. It can occur when brazing or welding copper components in a moist environment or within a reducing atmosphere containing hydrogen. In commercial copper grades which usually contain 200-600 ppm (0.02-0.06%) of oxygen, the result is a completely brittle metal. In Luvata OF-OK® copper, we guarantee a maximum of 5 ppm (0.0005%) oxygen content, which makes it immune to hydrogen embrittlement and ensures safe joining with all brazing and welding methods.

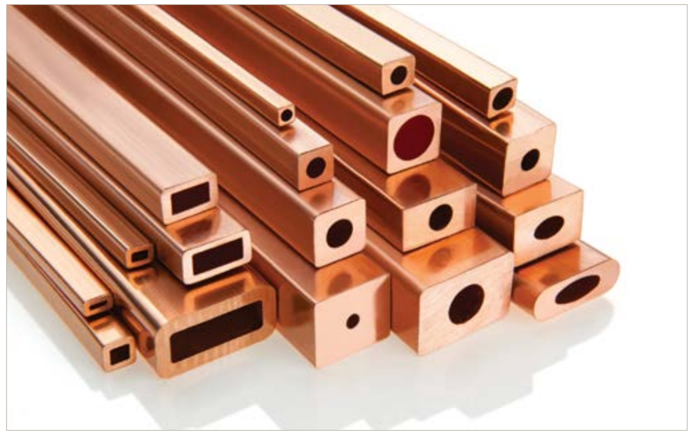
**Applications**
Luvata OF-OK® is the ideal material for a large number of applications:

- Rotor bars and stator winding for power generators
- Components for electrical equipment
- Coils for electrical magnets and induction furnaces
- Connectors
- Heat sinks for semiconductor bases

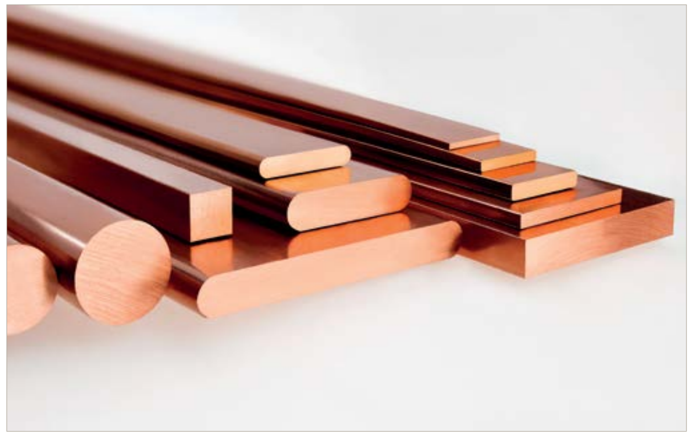


Hydrogen embrittlement in TIG welded, ETP (left) versus Luvata's TIG welding seam in oxygen-free OF-OK® copper (right)


XFEL corrector quadrupole



Luvata hollow conductors, OF-OK® and OFE-OK®



Luvata rods, bars and busbars, OF-OK® and OFE-OK®



# Luvata OFE-OK® Certified Grade Copper

## Luvata's OFE-OK® oxygen-free electronic grade copper is tested at every stage of the manufacturing process, from selecting the raw material to signing the inspection certification. That is why we call it Certified Grade Copper.

**A higher grade material for specialist applications**
For the most demanding electronic, cryogenic and optical applications, a higher grade oxygen-free copper is needed. Luvata OFE-OK® copper is a high-purity copper grade with total impurities of less than 40 ppm (0.0040%), with no single impurity exceeding 25 ppm (0.0025%). OFE-OK® copper is manufactured, controlled and inspected separately from standard Luvata OF-OK® copper. For the certified grade, only specially selected cathode copper is used and every ingot is inspected to ensure compliance with ASTM F68 Oxygen-Free Copper in Wrought Forms for Electron Devices specification and other international specifications for OFE copper.

**Vacuum Applications**
Luvata OFE-OK® Certified Grade Copper has an extremely low content of highly volatile low melting elements such as Phosphorus (P), Lead (Pb), Sulphur (S), Zinc (Zn), Cadmium (Cd) and Mercury (Hg). In a number of vacuum applications in the electronic industry, glass is directly bonded to copper, requiring an adherent oxide film on the copper surface. This oxide film can be rendered non-adherent and brittle if phosphorus, which is often used as oxidant, is present. To prevent this, Luvata OFE-OK® contains less than 3 ppm (0.0003%) of phosphorus, allowing for a direct bond to glass and ceramic materials. The low volatility and limited phosphorus content in Luvata OFE-OK® ensure the smooth functioning and long service life for vacuum components such as vacuum interrupters, X-ray tubes and radio transmitters.

**Integrity**
Luvata OFE-OK® is tested during all steps of the manufacturing process; casting, extrusion, drawing and at the end of the process to ensure the integrity of the material. Even small impurities or defects from extrusion may cause scattering of the beam from the optical component and cause rejection after a great deal of time and effort has been spent. Using OFE-OK® ensures the highest possible yield in the process. All Luvata OFE-OK® copper carries a test certificate guaranteeing the impurities are below the specified limits, electrical conductivity is at least 101.5% IACS and the material integrity complies with the requirements of the most demanding applications.

**Cryogenic Properties**
At cryogenic temperatures, the electrical and thermal properties of Luvata OFE-OK® copper are superior to any other copper grade. With the high purity of Luvata OFE-OK® combined with the special thermal treatment in the manufacturing process, the high value RRR (Residual Resistance Ratio) of Luvata OFE-OK® is guaranteed. RRR values up to 2000 are available with specific high purity copper grades.

Since RRR is very sensitive for common processing treatments, Luvata experts are pleased to advise the best thermomechanical treatment to meet the optimum RRR value in the final component. Unlike oxygen-containing copper alloys with high RRR values, Luvata OFE-OK® can readily be electron beam welded or annealed in a reducing atmosphere without the risk of hydrogen embrittlement.

On request, Luvata also provides copper materials having RRR within a specific range and alloyed copper grades with very low RRR value.


Laser mirrors


X-ray tube

# Technical data

**Alloys**

| Alloy | Luvata Alloy | Purity % | EN | ASTM |
|---|---|---|---|---|
| Cu-OF | OF-OK® | 99.95 min | CW008A | C10200 OF |
| Cu-OFE | OFE-OK® | 99.99 min | CW009A | C10100 OFE, ASTM F68 |
| CuAg0.04(OF) | HK003 | Cu + Ag min 99.95 | CW017A | C10400 OFS |
| CuAg0.10(OF) | HK015 | Cu + Ag min 99.95 | CW019A | C10700 OFS |

**Size range**

| Rods mm | Mill Length mm | Temper |
|---|---|---|
| 3 – 25 | 3500 – 4000 | hard, soft annealed |
| 25 – 60 | 3500 – 4000 | hard, soft annealed |
| 50 – 125 | 2000 – 3000 | hard, soft annealed |
| 125 – 190 | 1000 – 2000 | extruded, forged |
| 190 – 850 | 500 – 1500 | cast, forged |

**Busbars**

| Width mm | Min Thickness mm |
|---|---|
| 10 – 20 | 2 |
| 20 – 40 | 3 |
| 40 – 70 | 4 |
| 70 – 250 | 5 |

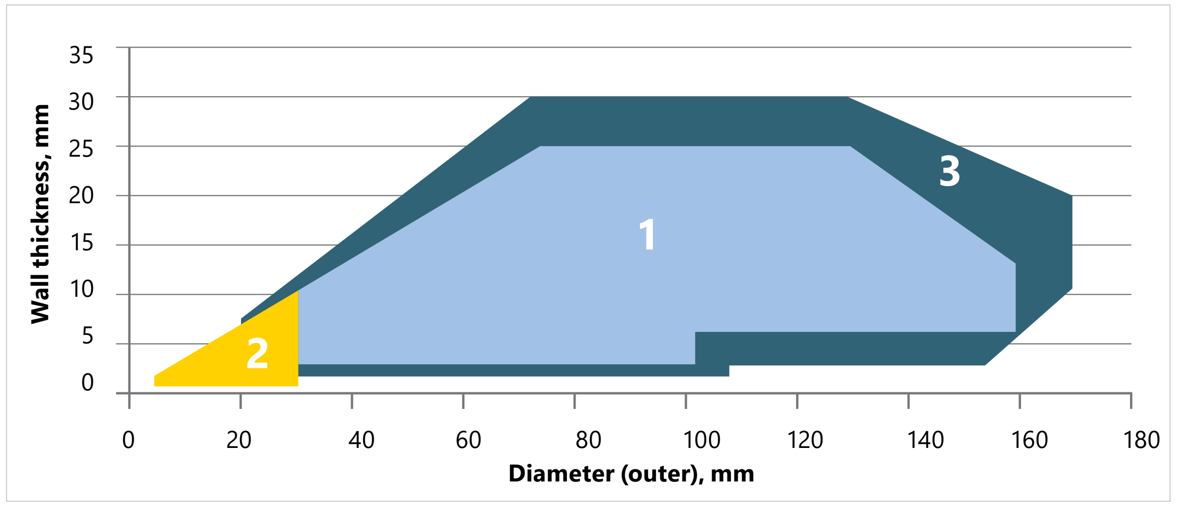
**Tubes**

**Deliver forms:**
- Straight lengths up to 12 m
- Loose coils up to 120 kg
- Mileon® continuous coils in wooden reels up to 2500 kg
- LWC (Level Wound Coil) coils
- Forged tubes up to OD 800 mm

**Hollow conductors**

**Size range:** Different shapes from 3 x 2 mm up to 100 x 100 mm.

**Delivery forms:**
- Straight lengths up to 16 m
- Loose coils
- Mileon® continuous coils in wooden reels up to 2500 kg
- Pancake coils
- LWC (Level Wound Coil) coils




See list of tools of profile tubes: **www.luvata.com/hollow-conductors**

Region No.1 – tubes in straight length.
Region No.2 – tubes in coil.
Region No.3 – special dimensions (available upon request). Other dimensions – check availability from Luvata.



# Luvata - Partnerships with a Promise

**Working for People, Society and the Earth**

We all want to ensure a sustainable future for our planet. As a manufacturer, we know that we have both an opportunity and a responsibility to contribute positively to the changes needed to secure that future. We have stated that our vision is to be the leading provider of copper solutions that benefit people, society and the earth, and this is a serious commitment. We are active in markets such as e-mobility, healthcare, electrification and the green transition, all of which have a need for copper products. We believe that our innovations will help drive these markets forwards, making green solutions more easily accessible to all.

We also recognize the need for our operations to be sustainable. To this end, we are striving to reduce greenhouse gas emissions, aiming to be using 100% renewable electricity by 2035 and to be fully carbon neutral by 2045.

**Helping our customers grow and thrive**

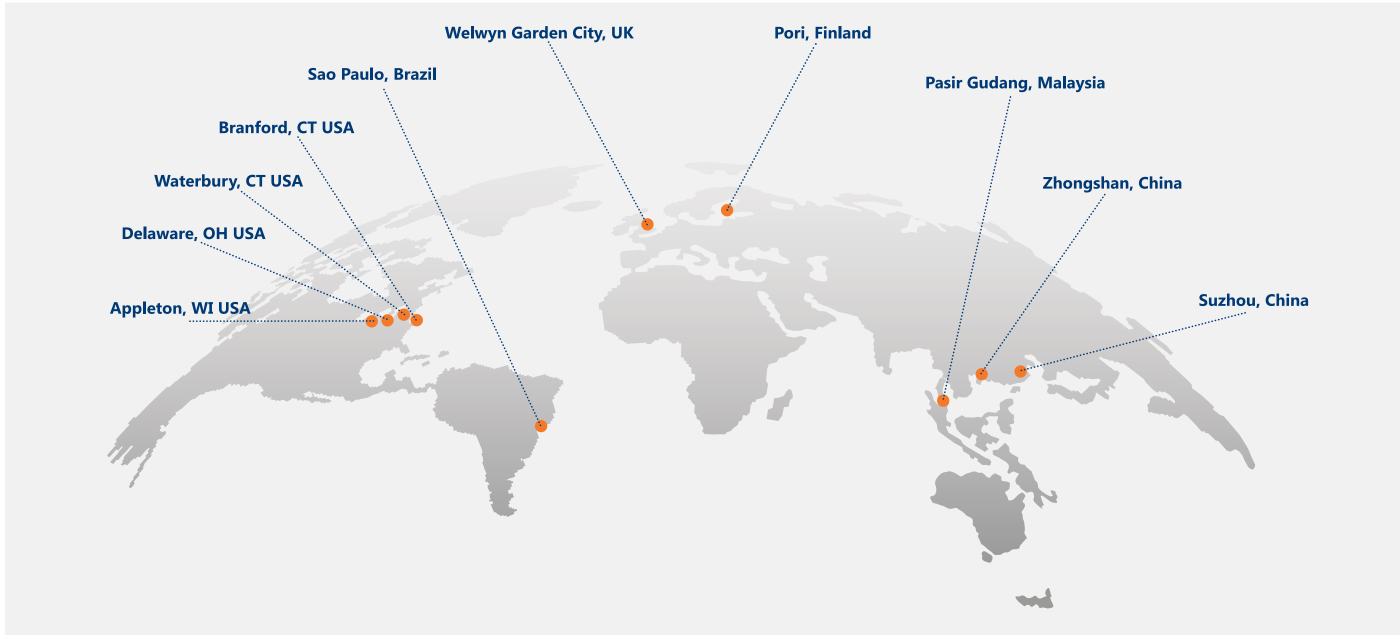
We have always worked in partnership with our customers, getting to know their businesses and collaborating on custom-made solutions. We take great satisfaction in seeing our customers thrive, and knowing that we played a part in their success.

Where we believe we can make a real difference is in sharing our robust manufacturing capabilities, technology and material knowledge, and by using our unique expertise to develop technically-demanding products. Together with an emphasis on lean production, innovation and value-added solutions, we work hard to ensure that we are the partner of choice for our customers, every time.

**Partnerships with a Promise**

At Luvata, we deliver on our promises. We have pledged always to treat people with respect, keep an open mindset and work together with others.

Our global footprint stretches across four continents. Our diversity of locations, cultures and markets gives us access to a wealth of knowledge and expertise that continues to grow. It also means that we are local to our customers, wherever they are, so we can continue to forge the strong relationships we are known for.





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**About Luvata**
Luvata is a world leader in copper manufacturing and related engineering services. It brings together people, innovation and technology to create solutions in areas such as e-mobility, healthcare, and in electrification & the green transition. The company's success is driven by innovation, lean production and value-added sales, and its vision is to provide copper solutions which benefit people, society and the earth. Employing over 1,400 staff on four continents, Luvata works in partnership with customers such as ABB, CERN and Siemens. Luvata is a group company of Mitsubishi Materials Corporation.

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# Oxygen-free Copper Cu-OF – Luvata Alloy OF-OK®

**Alloy description**

Luvata OF-OK® oxygen-free copper is high purity copper that is immune against hydrogen embrittlement. It is used in applications where high electrical and thermal conductivity are the essential requirements. It can be joined with all welding and brazing methods and it is suitable for manufacturing processes requiring extreme deformability.

**Typical applications:**

- Magnet windings
- Semiconductor components
- Electric motors
- Wave guide tubes
- Induction furnaces
- Electrical components
- Switchgear applications
- Generator material in rotor and stator windings
- Other applications where high electrical and thermal conductivity is needed

**Products / shapes:**

Profile tubes, round tubes, round rods, wire and strip coils, rectangular bars and solid profiles. Corresponding EN-norms for different products are as follows:

- EN 13600 – Copper and copper alloys.
  Seamless copper tubes for electrical purposes.
- EN 13601 – Copper and copper alloys.
  Copper rod, bar and wire for general electrical purposes.
- EN 13605 – Copper and copper alloys.
  Copper profiles and profiled wire for electrical purposes.

## Chemical composition and corresponding standards:

| Luvata Pori Oy alloy | Composition * % | EN – CEN/TS 13388:2008 | ASTM / USA |
|---|---|---|---|
| OF-OK | Cu + Ag min. 99,99 % | Cu-OF / CW008A | CDA C10200 |

* Other elements max %: O 0,0005,Bi 0,0005, Pb 0,005

## Physical properties:

| Density kg/dm$^3$ | Coefficient of linear expansion 1/K | Specific heat J/(kg x K) | Melting temperature °C |
|---|---|---|---|
| 8,94 | 0,0000177 | 385 | 1083 |

## Mechanical properties – typical values:

| | Soft temper | Half-hard temper | Hard temper |
|---|---|---|---|
| Hardness HV | 35 – 65 HV | 70 – 95 HV | 85 – 115 HV |
| Tensile strength | 200 – 220 N/mm$^2$ | 250 – 350 N/mm$^2$ | 260 – 400 N/mm$^2$ |
| 0,2% yield strength | 35 – 65 N/mm$^2$ | 180 – 280 N/mm$^2$ | 220 – 380 N/mm$^2$ |
| Elongation | min. 40 % | min. 12 % | min. 5 % |

## Electrical and thermal properties – typical values:

| | | | |
|---|---|---|---|
| Electrical conductivity | vol | % IACS * | min 100,6 |
| | mass | %IACS | min 100,0 |
| | MS/m | | min 58,3 |
| Electrical resistivity | vol | Ω mm$^2$/m | max 0,0171 |
| | mass | Ω g/m$^2$ | max 0,1532 |
| Thermal conductivity (20 °C) | W / Km | | 390 |

* % IACS = International Annealed Copper Standard. The % IACS values are calculated as percentages of the standard value for annealed high conductivity copper as laid down by the International Electrotechnical Commission.

## Joining and machining:

| Machinability rating (free cutting brass = 100) | Soldering | Brazing | TIG | MIG | EBW |
|---|---|---|---|---|---|
| 20 | Excellent | Excellent | Good | Good | Good |

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